2. Each ligand should have at least one filled orbital containing a lone pair of electrons.
3. In order to accommodate the electron pairs donated by the ligands, the central metal ion present in a complex provides required number (coordination number) of vacant orbitals.
4. These vacant orbitals of central metal atom undergo hybridisation, the process of mixing of atomic orbitals of comparable energy to form equal number of new orbitals called hybridised orbitals with same energy.
5. The vacant hybridised orbitals of the central metal ion, linearly overlap with filled orbitals of the ligands to form coordinate covalent sigma bonds between the metal and the ligand.
6. The hybridised orbitals are directional and their orientation in space gives a definite geometry to the complex ion.

| Coordination number | Hybridisation | Geometry | Examples |
|---|---|---|---|
| 2 | sp | Linear | $[CuCl_2]^-$, $[Ag(CN)_2]^-$ |
| 3 | $sp^2$ | Trigonal planar | $[HgI_3]^-$ |
| 4 | $sp^3$ | Tetrahedral | $[Ni(CO)_4]$, $[NiCl_4]^-$ |
| 4 | $dsp^2$ | Square planar | $[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$ |
| 5 | $dsp^3$ *($d_{x^2-y^2}$ orbital is involved)* | Trigonal bipyramidal | $Fe(CO)_5$ |
| 6 | $d^2sp^3$ *($d_{z^2}$ and $d_{x^2-y^2}$ orbitals of inner shell are involved)* | Octahedral | $[Ti(H_2O)_6]^{3+}$, $[Fe(CN)6]^{2-}$, $[Fe(CN)6]^{3-}$, $[Co(NH_3)_6]^{3+}$ *(Inner orbital complexes)* |
| 6 | $sp^3d^2$ *($d_{z^2}$ and $d_{x^2-y^2}$ orbitals of the outer shell are involved)* | Octahedral | $[FeF_6]^{4-}$,$[CoF_6]^{4-}$, $[Fe(H_2O)_6]^{2+}$ *(Outer orbital complexes)* |

7. In the octahedral complexes, if the (n-1) d orbitals are involved in hybridisation, then they are called inner orbital complexes or low spin complexes or spin paired complexes. If the nd orbitals are involved in hybridisation, then such complexes are called outer orbital or high spin or spin free complexes. Here n represents the principle quantum number of the outermost shell.
8. The complexes containing a central meal atom with unpaired electron(s) are paramagnetic. If all the electrons are paired, then the complexes will be diamagnetic.



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